

TEACH YOURSELF LECTURE

Lecture 01

Computer Systems

Chapter lecture — 20 topics with explanations, worked examples, Urdu notes, practice questions and a full answer key.

Computer Science XI · Bilingual support: English + اردو

کمپیوٹر سائنس — گیارہویں جماعت: خود سیکھیں، خود پرکھیں

20 teach-yourself lectures in this pack

★ Golden topics, Urdu notes, review and answer key

XI

How to use this lecture

Learn this chapter **on your own**. Read *Learn it*, try the worked example on paper, then answer *Check yourself* before looking at the answers. The green box repeats the idea in Urdu. ★ Golden topics are high-yield in board exams.

یہ لیکچر خود سیکھیں۔ مثال کاغذ پر حل کریں، پھر جواب دیکھنے سے پہلے خود کو پرکھیں۔

Lectures in this pack

Chapter 1: Computer Systems. Study one lecture at a time, then use the chapter review and answer key at the end of this PDF.

L01. 1.1.1 Discrete and Continuous Quantities

L03. 1.1.3 Analog and Digital Signals

L05. 1.1.5 Boolean Operations

L07. 1.1.7 Logic Gates (Basic Gates: AND, OR, NOT)

L09. 1.1.7 Advanced Logic Gates (XOR and XNOR)

L11. 1.1.11 Logic Diagrams

L13. LogiSim Evolution: Simulation Tool

L15. 1.2.1 – 1.2.3 Software Development Life Cycle (SDLC)

L17. 1.2.4 The Agile Model

L19. 1.3.1 – 1.3.4 Communication Models and the OSI 7-Layer Model

L02. 1.1.2 Introduction to Digital Systems

L04. 1.1.4 Boolean Algebra

L06. 1.1.6 Truth Tables

L08. 1.1.7 Universal Logic Gates (NAND and NOR)

L10. 1.1.8 – 1.1.10 Logic Expressions, Minterms and Maxterms

L12. 1.1.13 Karnaugh Maps (K-Maps)

L14. Building a Circuit in LogiSim

L16. 1.2.4 The Waterfall Model

L18. 1.2.5 – 1.2.7 Waterfall vs Agile: Comparison and Case Studies

L20. 1.3.5 – 1.3.6 TCP/IP Model vs OSI Model

CHAPTER 1

Computer Systems

This chapter shows you how a computer thinks in 0s and 1s, how logic gates turn those bits into decisions, how software is planned and built step by step, and how data travels across a network. These three ideas — digital logic, software development and communication models — are the foundation of everything else in Computer Science.

اس باب میں آپ سیکھیں گے کہ کمپیوٹر صفر اور ایک کی زبان میں کیسے کام کرتا ہے، لاجک گیٹس فیصلے کیسے کرتے ہیں، سافٹ ویئر مرحلہ وار کیسے بنایا جاتا ہے، اور نیٹ ورک پر ڈیٹا ایک جگہ سے دوسری جگہ کیسے پہنچتا ہے۔

What you will learn

- Tell the difference between discrete and continuous quantities, and between analog and digital signals, with examples.
- Use the Boolean operations AND, OR and NOT, and apply the order of precedence.
- Draw the symbol, write the expression and give the truth table of AND, OR, NOT, NAND, NOR, XOR and XNOR gates.
- Simplify Boolean expressions using 2-variable and 3-variable Karnaugh maps.
- Describe the six phases of the SDLC and compare the Waterfall and Agile models using case studies.
- State the advantages of digital systems.
- Build a complete truth table for any expression with 2 or 3 inputs.
- Write minterms (SOP) and maxterms (POS) from a truth table, and draw a logic diagram from an expression.
- Build and test a simple circuit in LogiSim Evolution.
- Explain the seven layers of the OSI model and map them to the four layers of the TCP/IP model.

1.1.1 Discrete and Continuous Quantities

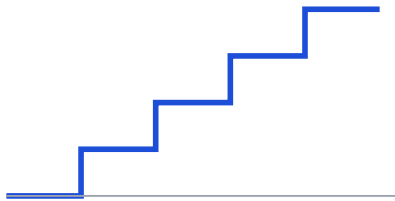
LEARN IT

Everything around you that can be measured or counted is a **quantity**. Quantities come in two kinds.

Discrete quantities are separate, countable items with specific values. You can count them one by one: 1, 2, 3 ... There is nothing "in between". You cannot have 2.5 students in a classroom.

Continuous quantities can take *any* value within a range. They flow smoothly without breaks. Water can be 30 °C, 30.5 °C, 30.51 °C, and so on — there is always another value in between.

Discrete (counted)	Continuous (measured)
Number of students in a classroom	Temperature of water
Number of cars in a parking lot	Speed of a moving car
Has fixed, separate values	Has any value in a range



Discrete (Stairs)



Continuous (Ramp)

Stairs have fixed steps (discrete). A ramp lets you stand at any point (continuous).

REMEMBER IT

Stairs vs Ramp. On stairs you can only stand on step 1, 2 or 3 — that is discrete data. On a ramp you can stand at any height — that is continuous data. Ask yourself: "Do I *count* it or *measure* it?" Count → discrete. Measure → continuous.

COMMON MISTAKE

Students sometimes say "money is continuous because amounts are large". No — rupees and paise are counted in fixed units, so money is discrete. Size does not matter; what matters is whether values in between are possible.

اردو میں سمجھیں

منفصل (Discrete) مقداریں وہ ہیں جنہیں ایک ایک کر کے گنا جا سکے، جیسے کلاس میں طلبہ کی تعداد۔ مسلسل (Continuous) مقداریں کسی حد کے اندر کوئی بھی قیمت لے سکتی ہیں، جیسے پانی کا درجہ حرارت۔ سیڑھیاں منفصل ڈیٹا کی مثال ہیں اور ڈھلوان (ریمپ) مسلسل ڈیٹا کی۔

CHECK YOURSELF

1. Is the number of pages in a book discrete or continuous?
2. Is your height discrete or continuous?
3. In the stairs/ramp analogy, which one shows continuous data?

Answers: 1. Discrete (you count pages). 2. Continuous (height is measured and can be any value). 3. The ramp.

1.1.2 Introduction to Digital Systems

LEARN IT

A **digital system** is an electronic system that uses only two values: **0** and **1**. These two values are called **binary digits**, or **bits** for short.

- **0** represents OFF, LOW, or FALSE.
- **1** represents ON, HIGH, or TRUE.

Why only two values? Because it is easy for an electronic circuit to tell "voltage present" from "no voltage". Every number, letter, photo and song in a computer is finally stored as a long pattern of these bits.

Advantages of digital systems

Advantage	Detail
Reliable	Less affected by noise and interference.
Accurate	No loss of quality when copying data.
Easy to design	Only two states (0 and 1) to handle.
Programmable	Can be easily controlled by software.

Examples of digital systems

- **Computers:** process data as 0s and 1s.
- **Smartphones:** store photos and music as binary data.
- **Digital cameras:** capture images as binary code.

WORKED EXAMPLE

Why is copying digital data "accurate"?

Step 1: A song is stored as bits, for example 10110010...

Step 2: When you copy it to a USB, the computer copies each bit exactly: 1 stays 1, 0 stays 0.

Step 3: Even after 100 copies the bits are identical, so the song sounds the same. An old cassette tape (analog) loses a little quality every time it is copied.

REMEMBER IT

Think of a bit as a light switch: it is either ON (1) or OFF (0). A digital system is a huge panel of such switches.

اردو میں سمجھیں

ڈیجیٹل نظام صرف دو قیمتیں استعمال کرتا ہے: 0 اور 1، جنہیں بیت (Bit) کہتے ہیں۔ 0 کا مطلب بند، کم یا غلط ہے اور 1 کا مطلب چالو، زیادہ یا درست ہے۔ ڈیجیٹل نظام قابل بھروسہ، درست، بنانے میں آسان اور سافٹ ویئر سے قابو میں رہنے والے ہوتے ہیں۔

CHECK YOURSELF

1. What does a 1 represent in a digital system?
2. Give two advantages of digital systems.

Answers: 1. ON, HIGH or TRUE. 2. Any two of: reliable (less affected by noise), accurate (no loss when copying), easy to design (two states), programmable (controlled by software).

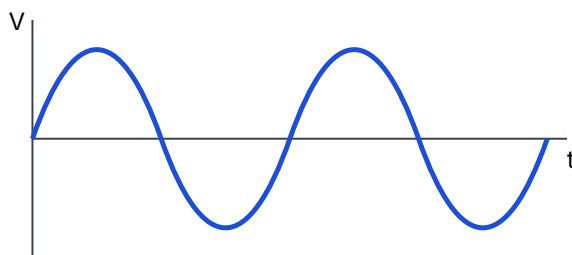
1.1.3 Analog and Digital Signals ★ Golden

LEARN IT

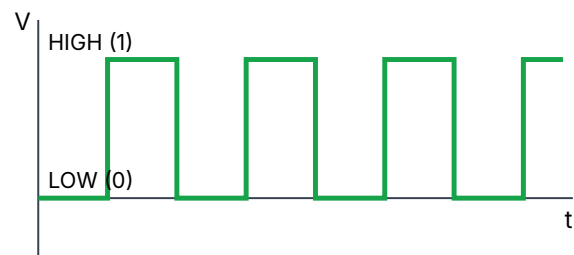
A **signal** is a changing voltage (or other quantity) that carries information.

Analog signals are continuous signals that change smoothly over time, like a **sine wave**. Examples: the human voice, a traditional mercury thermometer.

Digital signals have only two levels: **HIGH (1)** and **LOW (0)**. They jump instantly between these values, which creates a **square wave**. Example: computer data.



Analog Signal (Sine Wave)



Digital Signal (Square Wave)

An analog signal changes smoothly; a digital signal jumps between only two levels.

Feature	Analog data	Digital data
Nature	Continuous flow.	Discrete, separate values (bits).
Wave shape	Sine wave (smooth curve).	Square wave (sudden jumps).
Reliability	Low (affected by noise).	High (resistant to noise).
Examples	Human voice, traditional thermometer.	Computer data.

WORKED EXAMPLE

Why is digital more resistant to noise?

Step 1: Suppose a wire carries 5 V for HIGH and 0 V for LOW.

Step 2: Noise adds a small error, so 5 V arrives as 4.6 V and 0 V arrives as 0.3 V.

Step 3: The receiver only asks "Is it closer to HIGH or LOW?" 4.6 V is still read as 1 and 0.3 V is still read as 0. The data is perfect.

Step 4: In an analog signal, every exact value carries meaning, so 4.6 V instead of 5 V really changes the sound or reading. The noise cannot be removed.

REMEMBER IT

Analog = A smooth curve. Digital = Doorsteps. Analog looks like ocean waves; digital looks like a castle wall (square teeth).

COMMON MISTAKE

Do not draw a digital signal with sloping sides. A digital signal jumps *instantly*, so its edges are straight vertical lines. Also do not confuse "digital" with "electronic": an analog signal can also be electronic.

اینالوگ سگنل وقت کے ساتھ ہموار انداز میں بدلتا ہے، جیسے سائن ویو (Sine Wave)؛ انسانی آواز اس کی مثال ہے۔ ڈیجیٹل سگنل کی صرف دو سطحیں ہوتی ہیں، (1) HIGH اور (0) LOW، اور یہ اچانک ایک سے دوسری پر جاتا ہے، اس لیے مربع لہر (Square Wave) بنتی ہے۔ ڈیجیٹل سگنل شور سے کم متاثر ہوتا ہے، اس لیے زیادہ قابلِ بھروسہ ہے۔

EXAM TIP

Common questions: "Differentiate between analog and digital signals" (3–4 marks) and "Draw an analog and a digital signal." Write the comparison as a table with at least three features (nature, wave shape, reliability, example). Always label your diagrams: axes V and t, and HIGH(1)/LOW(0) on the square wave.

CHECK YOURSELF

1. What shape of wave does a digital signal make?
2. Which type of signal is more affected by noise?
3. Give one example of an analog signal.

Answers: 1. A square wave. 2. Analog. 3. Human voice (or a traditional thermometer reading).

1.1.4 Boolean Algebra

LEARN IT

Boolean algebra was developed by the English mathematician **George Boole**. It is a kind of algebra that deals with only two values: **TRUE (1)** and **FALSE (0)**. It is the foundation of digital electronics and computer programming — every **if** condition in a program is a small piece of Boolean algebra.

Concept	Description	Example
Variables	Letters that hold the value 0 or 1.	A, B, C, X, Y
Operations	Actions performed on variables.	AND, OR, NOT
Expressions	Combinations of variables and operations.	A AND B, $Y = A + B$

Some simple rules (laws) follow directly from the meaning of AND, OR and NOT. They are useful when you check or simplify answers:

OR rules	AND rules	NOT rules
$A + 0 = A$	$A \cdot 1 = A$	$(A')' = A$
$A + 1 = 1$	$A \cdot 0 = 0$	$A + A' = 1$
$A + A = A$	$A \cdot A = A$	$A \cdot A' = 0$

WORKED EXAMPLE

Using a law: Simplify $Y = A'B + AB$.

Step 1: Both terms contain B, so take B out: $Y = B(A' + A)$.

Step 2: $A' + A = 1$ (a variable OR its opposite is always 1).

Step 3: $Y = B \cdot 1 = B$.

COMMON MISTAKE

In Boolean algebra "+" means OR, not addition. So $1 + 1 = 1$ (TRUE OR TRUE is TRUE), not 2. There is no value bigger than 1.

اردو میں سمجھیں

بولین الجبرا جارج بول (George Boole) نے بنایا۔ اس میں صرف دو قیمتیں ہوتی ہیں: درست (1) اور غلط (0)۔ متغیرات (Variables) جیسے A اور B صرف 0 یا 1 رکھتے ہیں، آپریشنز AND، OR اور NOT ہیں، اور ان کے ملاپ سے ایکسپریشن (Expression) بنتا ہے۔

CHECK YOURSELF

1. Who developed Boolean algebra?
2. In Boolean algebra, what is $1 + 1$?

Answers: 1. George Boole. 2. 1 (because + means OR).

1.1.5 Boolean Operations

LEARN IT

There are three basic Boolean operations. Every other operation can be made from them.

1. **AND operation ($\cdot$)** — gives output 1 **only when ALL inputs are 1**. Expression: $Y = A \cdot B$ (often written just AB).
2. **OR operation ($+$)** — gives output 1 **when ANY input is 1**. Expression: $Y = A + B$.
3. **NOT operation (' or $\bar{}$)** — **inverts** (flips) the input. If the input is 1 the output is 0, and if the input is 0 the output is 1. It is a **unary** operation, meaning it works on only one operand. Expression: $Y = A'$ (also written as A with a bar on top, $\bar{A}$).

Operation	Real-life analogy
AND	A door with two locks. The door opens only when both locks are unlocked.
OR	A room with two doors. You can enter if either door is open.
NOT	A room light switch. When the switch is ON the room is lit; when it is OFF the room is dark — the state is the opposite of "dark".

WORKED EXAMPLE

Evaluate each expression for $A = 1$, $B = 0$.

- (a) $A \cdot B = 1 \cdot 0 = 0$ (not all inputs are 1).
- (b) $A + B = 1 + 0 = 1$ (at least one input is 1).
- (c) $B' = 0' = 1$ (NOT flips 0 to 1).
- (d) $A \cdot B' = 1 \cdot 1 = 1$.

REMEMBER IT

AND is "strict" — everyone must agree (all 1s). OR is "easy-going" — one 1 is enough. NOT is "the rebel" — it always says the opposite.

اردو میں سمجھیں

AND آپریشن تب ہی 1 دیتا ہے جب تمام ان پٹ 1 ہوں (دو تالوں والا دروازہ). OR آپریشن تب 1 دیتا ہے جب کوئی ایک ان پٹ بھی 1 ہو (دو دروازوں والا کمرہ). NOT آپریشن ان پٹ کو الٹ دیتا ہے اور صرف ایک ان پٹ پر کام کرتا ہے، اس لیے اسے یونری (Unary) کہتے ہیں۔

CHECK YOURSELF

1. What symbol is used for the OR operation?
2. Why is NOT called a unary operation?
3. Find $A + B'$ when $A = 0$, $B = 1$.

Answers: 1. "+". 2. Because it works on only one operand (one input). 3. $B' = 0$, so $0 + 0 = 0$.

1.1.6 Truth Tables

LEARN IT

A **truth table** shows **all possible input combinations** and the output for each one, for a logic gate or a circuit. It is the "complete record" of how the circuit behaves.

How to create a truth table:

1. Count the number of inputs (n).
2. Calculate total rows = 2^n .
3. List all input combinations in binary counting order (0, 1, 2, 3 ... in binary).
4. Calculate the output for each combination.

2-input example ($n = 2$, rows = 4)

A	B	Combination
0	0	Row 1
0	1	Row 2
1	0	Row 3
1	1	Row 4

3-input example ($n = 3$, rows = 8)

A	B	C	Decimal
0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7

REMEMBER IT

Quick fill trick for 3 inputs: In column C write 0,1,0,1... (change every row). In column B write 0,0,1,1... (change every 2 rows). In column A write 0,0,0,0,1,1,1,1 (change every 4 rows). For n inputs the leftmost column changes every 2^{n-1} rows.

WORKED EXAMPLE

Make the truth table of a 3-input AND: $Y = A \cdot B \cdot C$.

Step 1: $n = 3$, so rows = $2^3 = 8$.

Step 2: List the 8 combinations in binary order.

Step 3: AND gives 1 only when all inputs are 1, so only row 111 gives 1.

A	B	C	$Y = A \cdot B \cdot C$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

COMMON MISTAKE

Do not list rows in a random order — you may miss or repeat a combination. Always count in binary from 000 to 111. Also remember rows = 2^n , not $2 \times n$: 3 inputs give 8 rows, not 6.

اردو میں سمجھیں

ٹروٹھ ٹیبل (Truth Table) کسی گیٹ یا سرکٹ کے تمام ممکنہ ان پٹ اور ان کے آؤٹ پٹ دکھاتا ہے۔ اگر ان پٹ n ہوں تو قطاریں (Rows) 2 کی طاقت n ہوں گی؛ دو ان پٹ کے لیے 4 اور تین ان پٹ کے لیے 8 قطاریں۔ ان پٹ ہمیشہ بائری ترتیب میں لکھیں۔

CHECK YOURSELF

1. How many rows does a truth table with 4 inputs have?
2. In a 3-input table, what is the 6th row (counting from row 1 = 000)?

Answers: 1. $2^4 = 16$. 2. 101.

1.1.7 Logic Gates (Basic Gates: AND, OR, NOT) ★ Golden

LEARN IT

Logic gates are electronic circuits that perform Boolean operations. They take one or more input bits and produce one output bit. They are the **building blocks of all digital systems** — a processor contains billions of them.

The three basic gates match the three basic operations: the **AND gate**, the **OR gate** and the **NOT gate** (also called an **inverter**).



$$\text{AND: } Y = A \cdot B$$



$$\text{OR: } Y = A + B$$



$$\text{NOT: } Y = A'$$

Symbols of the basic gates. AND has a flat back and round front ("D" shape); OR has a curved back and pointed front; NOT is a triangle with a small circle (bubble).

AND gate — $Y = A \cdot B$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR gate — $Y = A + B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT gate — $Y = A'$

A	Y
0	1
1	0

WORKED EXAMPLE

A school bell circuit. The bell (Y) should ring when the timer says "period over" ($A = 1$) **and** the school is open ($B = 1$).

Step 1: The bell needs both conditions, so use an AND gate: $Y = A \cdot B$.

Step 2: Sunday, timer fires: $A = 1$, $B = 0 \rightarrow Y = 1 \cdot 0 = 0$. No bell.

Step 3: Monday, timer fires: $A = 1$, $B = 1 \rightarrow Y = 1$. The bell rings.

If instead the bell should ring when *either* the timer fires *or* the principal presses a button, you would use an OR gate.

REMEMBER IT

Shortcut: AND output is 0 if *any* input is 0. OR output is 1 if *any* input is 1. Learn just the "special row": AND is 1 only at 11; OR is 0 only at 00.

COMMON MISTAKE

Do not mix up the AND and OR symbols. AND has a **flat** input side (like the letter D). OR has a **curved** input side and a **pointed** output, like a shield or arrowhead.

اردو میں سمجھیں

لاچک گیٹس الیکٹرانک سرکٹ ہیں جو بولین آپریشن انجام دیتے ہیں اور تمام ڈیجیٹل نظاموں کی بنیادی اینٹیں ہیں۔ AND گیٹ صرف تب 1 دیتا ہے جب دونوں ان پٹ 1 ہوں، OR گیٹ صرف تب 0 دیتا ہے جب دونوں ان پٹ 0 ہوں، اور NOT گیٹ ان پٹ کو الٹ دیتا ہے۔

EXAM TIP

You are often asked to "draw the symbol, write the expression and truth table" of a gate (2–3 marks each). Draw neat symbols with labelled inputs A, B and output Y. Write the full truth table in binary order. A frequent MCQ: "Which gate gives 1 only when all inputs are 1?" (AND).

CHECK YOURSELF

1. What is the output of an OR gate when $A = 0$ and $B = 0$?
2. What is another name for the NOT gate?
3. For which input does a 2-input AND gate output 1?

Answers: 1. 0. 2. Inverter. 3. Only $A = 1$, $B = 1$.

1.1.7 Universal Logic Gates (NAND and NOR)

LEARN IT

NAND means "NOT AND": it is an AND gate followed by a NOT. **NOR** means "NOT OR": an OR gate followed by a NOT. On the symbol, the NOT is shown as a small bubble at the output.

NAND and NOR are called **universal gates** because **any logic circuit in the world can be built using only NAND gates (or only NOR gates)**. Factories like this: they can mass-produce one type of chip and build everything from it.



$$\text{NAND: } Y = (A \cdot B)'$$



$$\text{NOR: } Y = (A + B)'$$

NAND = AND symbol + bubble. NOR = OR symbol + bubble.

NAND (NOT AND) — $Y = (A \cdot B)'$

A	B	A·B	Y
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

NOR (NOT OR) — $Y = (A + B)'$

A	B	A+B	Y
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

WORKED EXAMPLE

Show that a NAND gate can work as a NOT gate.

Step 1: Connect the same input A to both inputs of the NAND gate. Then $Y = (A \cdot A)'$.

Step 2: Since $A \cdot A = A$, we get $Y = A'$.

Step 3: Check: $A = 0 \rightarrow (0 \cdot 0)' = 0' = 1$. $A = 1 \rightarrow (1 \cdot 1)' = 1' = 0$. This is exactly the NOT truth table.

In the same way, a NOR gate with both inputs tied together gives $(A + A)' = A'$. And a NAND followed by a NAND-as-NOT gives $((A \cdot B)')' = A \cdot B$, an AND gate. This is why they are "universal".

REMEMBER IT

NAND is the exact **opposite** of AND (0 only at 11). NOR is the exact opposite of OR (1 only at 00). Just flip the AND/OR output column.

COMMON MISTAKE

Do not forget the bubble. Without the small circle, a NAND symbol becomes an AND symbol and your answer is wrong. Also, $(A \cdot B)'$ is **not** the same as $A' \cdot B'$: for $A = 1, B = 0$, $(1 \cdot 0)' = 1$ but $0 \cdot 1 = 0$.

اردو میں سمجھیں

NAND کا مطلب ہے NOT AND اور NOR کا مطلب ہے NOT OR. انہیں یونیورسل (عالمگیر) گیٹ کہتے ہیں کیونکہ صرف NAND یا صرف NOR گیٹس سے دنیا کا کوئی بھی لاجک سرکٹ بنایا جا سکتا ہے۔ NAND صرف تب 0 دیتا ہے جب دونوں ان پٹ 1 ہوں، اور NOR صرف تب 1 دیتا ہے جب دونوں ان پٹ 0 ہوں۔

CHECK YOURSELF

1. Why are NAND and NOR called universal gates?
2. What is the NOR output for $A = 0$, $B = 1$?
3. How do you make a NOT gate from a NAND gate?

Answers: 1. Any logic circuit can be built using only NAND gates or only NOR gates. 2. 0. 3. Join both inputs together and feed A into them: $(A \cdot A)' = A'$.

1.1.7 Advanced Logic Gates (XOR and XNOR)

LEARN IT

XOR (Exclusive OR) gives output 1 when the inputs are **different**. Expression: $Y = A \oplus B$. In terms of basic operations, $A \oplus B = A'B + AB'$.

XNOR (Exclusive NOR) is XOR followed by NOT. It gives output 1 when the inputs are the **same**. Expression: $Y = (A \oplus B)'$ $= A'B' + AB$.

The XOR symbol is an OR symbol with an extra curved line at the input side. XNOR adds a bubble at the output.



$$\text{XOR: } Y = A \oplus B$$



$$\text{XNOR: } Y = (A \oplus B)'$$

XOR = OR symbol with an extra curve at the back. XNOR = XOR symbol + bubble.

$$\text{XOR — } Y = A \oplus B$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

$$\text{XNOR — } Y = (A \oplus B)'$$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

All seven gates at a glance

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

(NOT has one input: $A = 0 \rightarrow 1$, $A = 1 \rightarrow 0$.)

WORKED EXAMPLE

A staircase light. A light has one switch at the bottom (A) and one at the top (B) of a staircase. Flipping either switch should change the light.

Step 1: Start with both switches down: $A = 0$, $B = 0$, light off ($Y = 0$).

Step 2: Flip the bottom switch: $A = 1$, $B = 0 \rightarrow$ light on ($Y = 1$).

Step 3: Walk up and flip the top switch: $A = 1$, $B = 1 \rightarrow$ light off ($Y = 0$).

Step 4: The pattern 0, 1, 1, 0 is the XOR truth table. So the light is $Y = A \oplus B$.

REMEMBER IT

Memory trick: XOR outputs 1 when inputs are "eXclusively" different. XNOR outputs 1 when inputs are the **same**. XNOR is sometimes called the "equality gate".

COMMON MISTAKE

XOR is not the same as OR. They differ only in the last row: for $A = 1, B = 1$, OR gives 1 but XOR gives 0.

اردو میں سمجھیں

XOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ مختلف ہوں، اور XNOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ ایک جیسے ہوں۔ XNOR دراصل XOR کے بعد NOT ہے۔ OR اور XOR میں فرق صرف آخری قطار میں ہے جہاں دونوں ان پٹ 1 ہوں۔

CHECK YOURSELF

1. What is $1 \oplus 1$?
2. Which gate outputs 1 when both inputs are the same?
3. Write XOR using AND, OR and NOT.

Answers: 1. 0. 2. XNOR. 3. $A \oplus B = A'B + AB'$.

1.1.8 – 1.1.10 Logic Expressions, Minterms and Maxterms

LEARN IT

A **logic expression** (Boolean expression) combines variables with operations, for example $Y = A + B' \cdot C$. To evaluate it correctly you must follow the **order of precedence**, just like BODMAS in maths:

1. Parentheses () → 2. NOT (') → 3. AND (·) → 4. OR (+)

WORKED EXAMPLE

Worked example 1: truth table of $Y = A + B' \cdot C$

Step 1: 3 inputs, so 8 rows.

Step 2: Precedence: first find B' (NOT), then $B' \cdot C$ (AND), then $A + (B' \cdot C)$ (OR). Make a column for each step.

A	B	C	B'	$B' \cdot C$	$Y = A + B' \cdot C$
0	0	0	1	0	0
0	0	1	1	1	1
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	1	1	1
1	1	0	0	0	1
1	1	1	0	0	1

Step 3: $Y = 1$ in rows 1, 4, 5, 6, 7 (decimal). Check one row by hand: $A=0, B=0, C=1 \rightarrow B' = 1, B' \cdot C = 1, Y = 0 + 1 = 1$.

✓

WORKED EXAMPLE

Worked example 2: truth table of $Y = (A + B) \cdot C'$

Step 1: Parentheses first: find $A + B$. Then NOT: C' . Then AND the two.

A	B	C	A+B	C'	Y
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	1	1	1
0	1	1	1	0	0
1	0	0	1	1	1
1	0	1	1	0	0
1	1	0	1	1	1
1	1	1	1	0	0

Step 2: $Y = 1$ in rows 2, 4 and 6 (decimal).

Minterms (SOP) and Maxterms (POS)

LEARN IT

Sometimes you are given a truth table and must *find* the expression. There are two standard ways.

Minterm (used in SOP — Sum of Products): a **product (AND)** of all the variables, written for a row whose output is **1**. A variable is **complemented (A')** if its value is 0, and **uncomplemented (A)** if its value is 1. OR-ing all the minterms gives the SOP expression.

Maxterm (used in POS — Product of Sums): a **sum (OR)** of all the variables, written for a row whose output is **0**. A variable is **complemented (A')** if its value is 1, and **uncomplemented (A)** if its value is 0. AND-ing all the maxterms gives the POS expression.

A	B	Output (Y)	Minterm (SOP)	Maxterm (POS)
0	0	0	—	$A + B$
0	1	1	$A'B$	—
1	0	0	—	$A' + B$
1	1	1	AB	—

Final SOP expression: $Y = A'B + AB$. **Final POS expression:** $Y = (A + B)(A' + B)$. Both describe the same table. (Both simplify to $Y = B$ — look at the table: Y always equals B . You will confirm this with a K-map later.)

WORKED EXAMPLE

Worked example 3: write the SOP for a 3-input "majority" circuit. $Y = 1$ when at least two of A , B , C are 1.

Row	A	B	C	Y	Minterm
0	0	0	0	0	–
1	0	0	1	0	–
2	0	1	0	0	–
3	0	1	1	1	A'BC
4	1	0	0	0	–
5	1	0	1	1	AB'C
6	1	1	0	1	ABC'
7	1	1	1	1	ABC

Step 1: Pick the rows with Y = 1: rows 3, 5, 6, 7.

Step 2: Row 3 is A=0, B=1, C=1 → A is 0 so write A'; B and C are 1 so write B, C → A'BC.

Step 3: Do the same for the other rows and OR them: **Y = A'BC + AB'C + ABC' + ABC**. In short form: **Y = Σm(3, 5, 6, 7)**. We will simplify this with a K-map in 1.1.13.

REMEMBER IT

SOP = 1s, POS = 0s. Minterm: 0 → bar, 1 → plain. Maxterm: the reverse (1 → bar, 0 → plain). Think "Min likes 1s, Max likes 0s".

COMMON MISTAKE

Two classic errors: (1) writing minterms for the 0-rows — minterms come only from rows where Y = 1; (2) forgetting precedence — in $A + B \cdot C$ you must do $B \cdot C$ first. $A + B \cdot C$ is **not** $(A + B) \cdot C$: for A=1, B=0, C=0 the first gives 1 and the second gives 0.

اردو میں سمجھیں

ایکسپریشن حل کرتے وقت ترتیب یہ ہے: پہلے قوسین، پھر NOT، پھر AND اور آخر میں OR. منترم (Minterm) اُن قطاروں سے بنتا ہے جن کا آؤٹ پٹ 1 ہو؛ اس میں 0 والے متغیر پر بار (A') لگتی ہے۔ میکس ٹرم (Maxterm) اُن قطاروں سے بنتا ہے جن کا آؤٹ پٹ 0 ہو؛ اس میں 1 والے متغیر پر بار لگتی ہے۔

CHECK YOURSELF

1. Which comes first: AND or OR?
2. Write the minterm for A = 1, B = 0, C = 1.
3. Write the maxterm for A = 0, B = 1.

Answers: 1. AND. 2. AB'C. 3. A + B'.

1.1.11 Logic Diagrams ★ Golden

LEARN IT

A **logic diagram** shows how logic gates are connected to create a circuit. It is a "picture" of a Boolean expression. To draw one, use the order of precedence: the operation with the **highest priority is drawn first** (nearest the inputs), and the last operation is the gate that gives the final output.

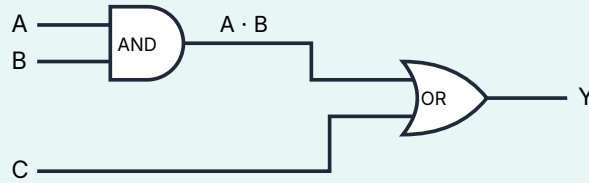
WORKED EXAMPLE

Example 1: Draw the logic diagram for $Y = A \cdot B + C$.

Step 1: Find the operations: AND ($\cdot$) and OR ($+$).

Step 2: AND has higher precedence than OR, so draw the AND gate first, with inputs A and B. Its output is $A \cdot B$.

Step 3: Draw an OR gate. Feed it $A \cdot B$ and C. Its output is Y.



$Y = A \cdot B + C$: AND first, then OR.

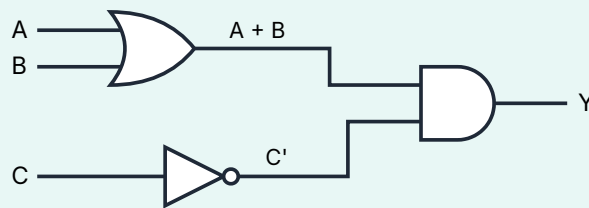
WORKED EXAMPLE

Example 2: Draw the logic diagram for $Y = (A + B) \cdot C'$.

Step 1: Parentheses have the highest priority, so draw the OR gate for $A + B$ first.

Step 2: NOT is next: draw a NOT gate on input C to get C' .

Step 3: The last operation is AND: feed $(A + B)$ and C' into an AND gate to get Y.



$Y = (A + B) \cdot C'$: OR and NOT first, then AND.

You can check this circuit against the truth table in 1.1.8 (worked example 2): $Y = 1$ only for rows 010, 100 and 110.

REMEMBER IT

Read the expression and ask "What is done **last**?" That gate goes on the far right, next to Y. Work backwards to the inputs. Count gates: $Y = A \cdot B + C$ needs 2 gates; $Y = (A+B) \cdot C'$ needs 3.

COMMON MISTAKE

Do not draw a separate wire for a variable each time it appears and then leave one unconnected. Every gate input must be joined to an input variable or to another gate's output, and every wire should be clearly labelled in exams.

اردو میں سمجھیں

لاجک ڈایاگرام دکھاتا ہے کہ گیٹس آپس میں کیسے جڑ کر سرکٹ بناتے ہیں۔ ترجیحی ترتیب کے مطابق جو آپریشن پہلے ہوتا ہے اس کا گیٹ پہلے (ان پٹ کے قریب) بنایا جاتا ہے، اور آخری آپریشن کا گیٹ آؤٹ پٹ Y دیتا ہے۔ مثلاً $Y = A \cdot B + C$ میں پہلے AND گیٹ اور پھر OR گیٹ بنتا ہے۔

EXAM TIP

Typical question: "Draw the logic diagram of $Y = \dots$ " and sometimes "also make its truth table" (4–5 marks). Write the precedence steps in one line, draw neat gate symbols with labelled intermediate outputs (like A·B), and finish with the truth table. Marks are lost for wrong gate shapes and missing bubbles.

CHECK YOURSELF

1. In $Y = A + B \cdot C$, which gate is drawn first?
2. How many gates are needed for $Y = A' \cdot B$?

Answers: 1. The AND gate (for $B \cdot C$). 2. Two: one NOT gate (for A') and one AND gate.

1.1.13 Karnaugh Maps (K-Maps) ★ Golden

LEARN IT

A **Karnaugh map (K-map)** is a graphical tool used to **simplify Boolean expressions**. It is a grid of cells — one cell for each row of the truth table. The cells are arranged so that **neighbouring (adjacent) cells differ by only ONE bit**. This ordering is called **Gray code**.

You write the output (1 or 0) of each row into its cell, then **group the 1s** in rectangles of size 1, 2, 4 or 8 (powers of 2). Each group becomes one simple term. This simplifies the equation without long algebra.

Two-variable K-map (A, B)

2 variables need $2^2 = 4$ cells.

	B = 0	B = 1
A = 0	Cell 0 ($A'B'$)	Cell 1 ($A'B$)
A = 1	Cell 2 (AB')	Cell 3 (AB)

Three-variable K-map (A, B, C)

3 variables need $2^3 = 8$ cells. The columns follow the Gray code sequence **00, 01, 11, 10** (not 00, 01, 10, 11). This makes adjacent cells differ by only one variable.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	m_0 ($A'B'C'$)	m_1 ($A'B'C$)	m_3 ($A'BC$)	m_2 ($A'BC'$)
A = 1	m_4 ($AB'C'$)	m_5 ($AB'C$)	m_7 (ABC)	m_6 (ABC')

LEARN IT

Grouping rules

1. Group only 1s. Groups must be rectangles of size 1, 2, 4 or 8 — never 3, 5 or 6.
2. Make groups **as large as possible**. Bigger groups give simpler terms.
3. Use **as few groups as possible**, but every 1 must be in at least one group.
4. **Overlapping is allowed** — a 1 may be used in more than one group.
5. **Wrapping around the edges is allowed**: a cell in the far-left column is adjacent to the cell in the far-right column (BC = 00 is next to BC = 10).
6. To read a group: variables that **change** (0 to 1) inside the group are **eliminated**. Variables that stay the same are kept — plain if they stay 1, complemented if they stay 0.

Group size tells you how many variables disappear: in a 3-variable map, a group of 2 leaves 2 variables, a group of 4 leaves 1 variable, and a group of 8 means $Y = 1$.

WORKED EXAMPLE

K-map example 1 (2 variables): simplify $Y = A'B + AB$.

Step 1: The minterms are $A'B$ (cell 1) and AB (cell 3). Put 1 in those cells, 0 elsewhere.

	B = 0	B = 1
A = 0	0	1
A = 1	0	1

Step 2: The two 1s are in the same column, one above the other. Make one group of 2.

Step 3: Inside the group, A changes (0 then 1) → eliminate A. B stays 1 → keep B.

Result: $Y = B$. This agrees with the algebra in 1.1.4 and the minterm table in 1.1.8.

WORKED EXAMPLE

K-map example 2 (2 variables, overlapping): simplify $Y = A'B' + A'B + AB$.

Step 1: Put 1s in cells 0, 1 and 3.

	B = 0	B = 1
A = 0	1	1
A = 1	0	1

Step 2: Group 1 = the top row (cells 0 and 1). A stays 0, B changes → term A' .

Step 3: Group 2 = the right column (cells 1 and 3). A changes, B stays 1 → term B . Cell 1 is used twice — overlapping is allowed.

Result: $Y = A' + B$. Check: $A' + B$ is 0 only when $A = 1$ and $B = 0$, which is exactly the one 0 cell (cell 2). ✓

WORKED EXAMPLE

K-map example 3 (3 variables): simplify the majority function $Y = \Sigma m(3, 5, 6, 7) = A'BC + AB'C + ABC' + ABC$.

Step 1: Fill the map. Careful — the columns are in Gray order 00, 01, 11, 10, so m_3 is in the third column and m_2 in the fourth.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	0	0	1 (m_3)	0
A = 1	0	1 (m_5)	1 (m_7)	1 (m_6)

Step 2: No group of 4 is possible (there is no 2×2 block or full row of 1s). So make groups of 2, each using m_7 :

- Group 1: $m_3 + m_7$ (column BC = 11). A changes; B = 1, C = 1 stay → BC .
- Group 2: $m_5 + m_7$ (row A = 1, columns 01 and 11). B changes; A = 1, C = 1 stay → AC .
- Group 3: $m_7 + m_6$ (row A = 1, columns 11 and 10). C changes; A = 1, B = 1 stay → AB .

Result: $Y = AB + BC + AC$. Four 3-variable terms (needing 4 three-input AND gates) became three 2-variable terms.

Check one row: $A=1, B=0, C=1 \rightarrow AB = 0, BC = 0, AC = 1 \rightarrow Y = 1$. In the truth table, row 5 is 1. ✓

WORKED EXAMPLE

K-map example 4 (3 variables, wrap-around): simplify $Y = \Sigma m(0, 2, 4, 6)$.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	1	0	0	1
A = 1	1	0	0	1

Step 1: The 1s are in the far-left column (BC = 00) and the far-right column (BC = 10).

Step 2: Because the map wraps around, these two columns are adjacent. Make **one group of 4**: m_0, m_4, m_2, m_6 .

Step 3: In this group A changes (0, 1) and B changes (0, 1). Only C stays the same: C = 0 $\rightarrow$ term **C'**.

Result: $Y = C'$. Without wrapping you would get two groups ($B'C' + BC'$), which is correct but not the simplest.

WORKED EXAMPLE

K-map example 5 (3 variables): simplify $Y = \Sigma m(1, 4, 5, 6, 7)$ — this is the output of $Y = A + B'C$ from 1.1.8.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	0	1	0	0
A = 1	1	1	1	1

Step 1: Group of 4: the whole row A = 1 (m_4, m_5, m_7, m_6). B and C both change; A stays 1 $\rightarrow$ **A**.

Step 2: m_1 is still alone. Pair it with m_5 below it (column BC = 01). A changes; B = 0, C = 1 stay $\rightarrow$ **B'C**.

Result: $Y = A + B'C$ — the K-map confirms the original expression was already in its simplest form.

WORKED EXAMPLE

Going further — a 4-variable K-map: simplify $Y = \Sigma m(0, 2, 5, 7, 8, 10, 13, 15)$. (Not required by the slides, but a good test of the same rules.) 4 variables need 16 cells; both rows (AB) and columns (CD) use Gray order 00, 01, 11, 10.

	CD = 00	CD = 01	CD = 11	CD = 10
AB = 00	1 (m_0)	0	0	1 (m_2)
AB = 01	0	1 (m_5)	1 (m_7)	0
AB = 11	0	1 (m_{13})	1 (m_{15})	0
AB = 10	1 (m_8)	0	0	1 (m_{10})

Step 1: The centre 2x2 block (m_5, m_7, m_{13}, m_{15}) is a group of 4. A and C change; B = 1 and D = 1 stay $\rightarrow$ **BD**.

Step 2: The four corners (m_0, m_2, m_8, m_{10}) are adjacent through wrap-around (left-right *and* top-bottom). A and C change; B = 0 and D = 0 stay $\rightarrow$ **B'D'**.

Result: $Y = BD + B'D'$ (this is B XNOR D). Eight 4-variable minterms became two 2-variable terms.

REMEMBER IT

Gray code song: "zero-zero, zero-one, one-one, one-zero" — only one bit flips at each step. Imagine the K-map is drawn on a tube, so the left and right edges touch. For reading groups: **"If it changes, it goes."**

COMMON MISTAKE

Most common K-map mistakes: (1) writing the columns as 00, 01, 10, 11 — this puts m_2 and m_3 in the wrong places; (2) making groups of 3 or 6 — only 1, 2, 4, 8 are allowed; (3) grouping diagonally — groups must be rectangles of side-by-side cells; (4) stopping at small groups when a bigger group (or a wrap-around group) is possible.

اردو میں سمجھیں

کارنو میپ (K-Map) بولین ایکسپریشن کو آسان بنانے کا تصویری طریقہ ہے۔ اس کے ساتھ والے خانوں میں صرف ایک بٹ کا فرق ہوتا ہے (گرے کوڈ: 00, 01, 10, 11)۔ 1 والے خانوں کو 1, 2, 4, 8 یا 8 کے گروپوں میں جتنا بڑا ممکن ہو جوڑیں؛ گروپ ایک دوسرے پر چڑھ سکتے ہیں اور کناروں سے گھوم کر بھی بن سکتے ہیں۔ گروپ کے اندر جو متغیر بدلتا ہے وہ ختم ہو جاتا ہے۔

EXAM TIP

K-maps are a favourite long question (5–6 marks): "Simplify $F(A,B,C) = \sum m(\dots)$ using a K-map." Draw the map with labelled Gray-code headings, fill every cell (write the 0s too), circle the groups clearly, write the term for each group beside it, and give the final simplified expression. Examiners often add "draw the logic diagram of the result" — so practise drawing it too.

CHECK YOURSELF

1. In a 3-variable K-map, which column is next to $BC = 01$ on the right?
2. Simplify $Y = \sum m(2, 3)$ for two variables A, B .
3. Simplify $Y = \sum m(4, 5, 6, 7)$ for three variables.

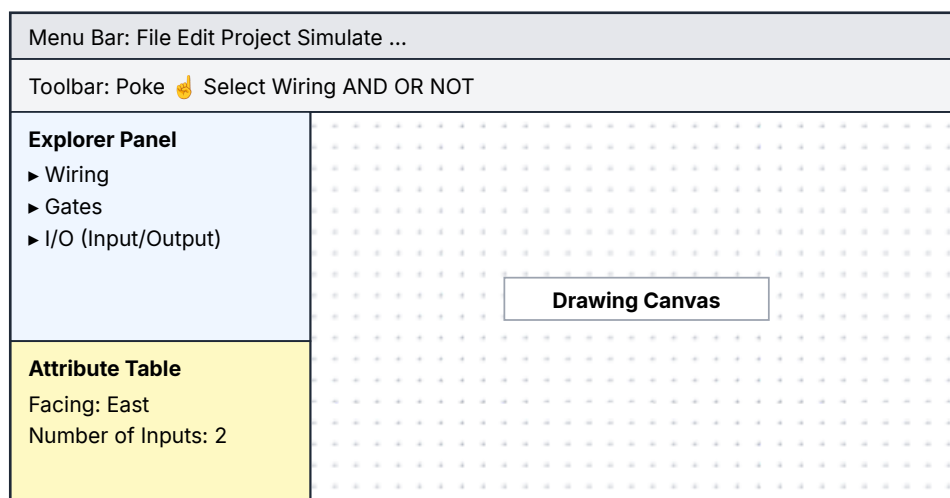
Answers: 1. $BC = 11$. 2. Cells 2 and 3 are the row $A = 1$; B changes $\rightarrow Y = A$. 3. The whole row $A = 1$ is a group of 4 $\rightarrow Y = A$.

LogiSim Evolution: Simulation Tool

LEARN IT

LogiSim Evolution is a free graphical tool for **designing and simulating logic circuits**. "Simulating" means the computer pretends to be the real circuit: you can switch inputs on and off and watch the output change. It lets you see the truth tables and Boolean expressions you learn in theory.

1. The interface areas



The main areas of the LogiSim Evolution window.

Area	What it does
Menu Bar (top)	File, Edit, Project, Simulate menus — open/save files and control the simulation.
Toolbar	Quick access to standard tools: Wiring, Poke, and AND, OR, NOT gates.
Explorer Panel (left)	Contains the component libraries: Gates, I/O, Wiring.
Attribute Table (bottom left)	Adjust the properties of the selected gate, e.g. Number of Inputs or Facing direction.
Drawing Canvas (right)	The dotted grid where you place and wire the gates.

Wiring colours during simulation

Colour	Meaning
Bright green	Logic 1 (HIGH)
Dark green	Logic 0 (LOW)
Blue	Unknown / unconnected pin
Red	Error / short circuit

REMEMBER IT

Colours like a traffic light: Green means "working" (bright = 1, dark = 0). Blue means "the wire is lonely — not connected". Red means "stop, something is wrong".

COMMON MISTAKE

If your output wire is blue, the circuit is not wrong in logic — a wire is simply not touching a pin. Zoom in and make sure every wire end sits exactly on a connection point.

اردو میں سمجھیں

لاجی سیم ایوولوشن (LogiSim Evolution) لاجک سرکٹ بنانے اور چلا کر دیکھنے کا سافٹ ویئر ہے۔ اس میں مینو بار، ٹول بار، ایکسپلورر پینل، ایٹری بیوٹ ٹیبل اور ڈرائنگ کینوس ہوتے ہیں۔ چمکدار سبز تار 1، گہری سبز 0، نیلی تار غیر منسلک اور سرخ تار غلطی کو ظاہر کرتی ہے۔

CHECK YOURSELF

1. Where do you change the number of inputs of a gate?
2. What does a red wire mean?

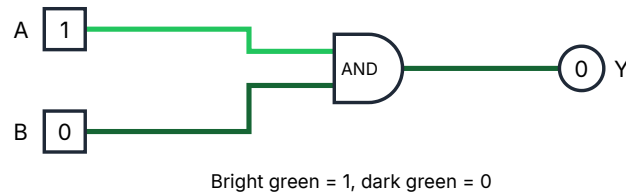
Answers: 1. In the Attribute Table (bottom left). 2. An error or short circuit.

Building a Circuit in LogiSim

WORKED EXAMPLE

Step by step: building $Y = A \cdot B$

1. Select the **AND gate** from the Toolbar (or from the Gates library) and place it on the canvas.
2. Select the **Input Pin** (square icon with a 0) and place two pins to the left of the gate. Label them A and B.
3. Select the **Output Pin** (circle icon with a 0) and place it to the right. Label it Y.
4. Use the **Wiring Tool** to click and drag wires from the input pins to the AND gate's input ports. Wires must be horizontal or vertical.
5. Wire the gate's output to the output pin Y.
6. Select the **Poke Tool** (pointing finger) and click inputs A and B to toggle them between 0 and 1. Test the circuit!



Testing with the Poke Tool: A = 1 (bright green wire), B = 0 (dark green), so $Y = 1 \cdot 0 = 0$.

WORKED EXAMPLE

Testing the circuit = making its truth table. Poke the inputs through all four combinations and write what Y shows:

Poke A	Poke B	Y shows	Output wire colour
0	0	0	Dark green
0	1	0	Dark green
1	0	0	Dark green
1	1	1	Bright green

This matches the AND truth table, so the circuit is correct. Try the same with $Y = A \cdot B + C$ from 1.1.11 (add a second gate and a third input pin).

REMEMBER IT

Input pin = **square** (you push it). Output pin = **circle** (it shows a light). Poke = finger — you "poke" inputs to change them.

COMMON MISTAKE

Students often try to poke inputs while the Wiring or Select tool is active — then clicking moves the pin instead of changing it. Switch to the Poke Tool first.

اردو میں سمجھیں

لاجی بسم میں $Y = A \cdot B$ بنانے کے لیے AND گیٹ رکھیں، بائیں جانب دو ان پٹ پن (چوکور) اور دائیں جانب ایک آؤٹ پٹ پن (گول) رکھیں، پھر وائرنگ ٹول سے تاریں جوڑیں۔ آخر میں پوک ٹول (انگلی) سے ان پٹ کو 0 اور 1 کر کے سرکٹ کو جانچیں۔

CHECK YOURSELF

1. Which icon is the input pin: square or circle?
2. Which tool do you use to change an input between 0 and 1?

Answers: 1. Square (output pin is the circle). 2. The Poke Tool (pointing finger).

1.2.1 – 1.2.3 Software Development Life Cycle (SDLC)

★ Golden

LEARN IT

The **Software Development Life Cycle (SDLC)** is a structured process that guides software development from beginning to end. It ensures **clear goals, better planning, risk reduction and high quality**. Without it, teams forget features, go over budget and deliver buggy software.

SDLC has **6 phases**:



Phases 1–3 in detail

Phase	Activities and details	Deliverable (what it produces)
1. Requirement Analysis	Meet clients, understand their needs, list the features. This is the most important phase — mistakes here spread to every later phase.	SRS Document (Software Requirements Specification)
2. System Design	Create the High-Level Design (overall architecture), the Low-Level Design (database, UI screens) and flowcharts.	Design Document
3. Implementation	Write the actual code in programming languages (Python, Java, etc.) based on the design documents.	Working Source Code

Phases 4–6 in detail

Phase	Activities and details	Deliverable
4. Testing	Find and fix bugs (errors), using Black Box and White Box testing.	Test Reports, bug-free software
5. Deployment	Release the software to end-users: Direct, Phased, or Pilot release.	Live Software
6. Maintenance	Provide support and updates, and fix post-release issues.	Patches, Updated Versions

Deployment methods: In a **direct** release the old system is switched off and the new one switched on at once. In a **phased** release the new system is introduced part by part. In a **pilot** release one small group (for example one branch or one class) uses it first before everyone else.

Testing methods

Black Box Testing

- The tester does **not** see the code.
- Tests functionality: give inputs, check outputs.
- Done by dedicated testers.

White Box Testing

- The tester **can** see the code.
- Tests the internal logic and structure.
- Done by developers.

WORKED EXAMPLE

Building a house = SDLC.

Phase 1 (Requirements): you ask the family what rooms they need — 3 bedrooms, a kitchen, a garage.

Phase 2 (Design): the architect draws the blueprints.

Phase 3 (Implementation): the builders lay the bricks.

Phase 4 (Testing): an inspector checks the wiring and the plumbing.

Phase 5 (Deployment): the family moves in.

Phase 6 (Maintenance): repairs, painting and a new room added later.

WORKED EXAMPLE

Black box vs white box on a login screen. A black-box tester types a correct password and a wrong password and checks that the first is accepted and the second rejected — without looking at the code. A white-box tester (the developer) reads the code and checks that every `if` branch, for example the "account locked after 3 tries" branch, actually runs correctly.

REMEMBER IT

Remember the order with "Real Developers Implement, Test, Deploy, Maintain" (R-D-I-T-D-M). Black box = a closed box (you can't see inside). White box = a glass box (you can see inside).

COMMON MISTAKE

Do not say the SDLC ends at deployment. Maintenance is a full phase and often lasts the longest. Also, the SRS is the deliverable of Requirement Analysis, not of Design.

اردو میں سمجھیں

ایس ڈی ایل سی (SDLC) سافٹ ویئر بنانے کا منظم طریقہ ہے جس کے چھ مراحل ہیں: ضروریات کا تجزیہ، سسٹم ڈیزائن، کوڈنگ (Implementation)، ٹیسٹنگ، تنصیب (Deployment) اور دیکھ بھال (Maintenance)۔ پہلا مرحلہ سب سے اہم ہے اور اس سے SRS دستاویز بنتی ہے۔ بلیک باکس ٹیسٹنگ میں کوڈ نظر نہیں آتا جبکہ وائٹ باکس ٹیسٹنگ میں کوڈ دیکھ کر اندرونی منطق جانچی جاتی ہے۔

EXAM TIP

"Explain the phases of SDLC" is a very common long question (5–8 marks). Name all 6 phases in order, write 1–2 lines of activities for each, and state the deliverable of each (SRS, Design Document, Source Code, Test Reports, Live Software, Patches). A short question often asks you to differentiate black box and white box testing — answer in a 3-row table (see code?, what is tested?, who does it?).

CHECK YOURSELF

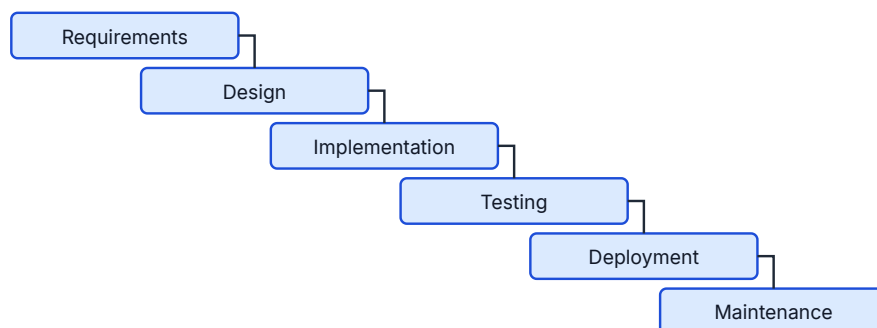
1. Which phase produces the SRS document?
2. Who usually performs white box testing?
3. Name the three ways to deploy software.

Answers: 1. Requirement Analysis. 2. Developers. 3. Direct, Phased and Pilot release.

1.2.4 The Waterfall Model ★ Golden

LEARN IT

The **Waterfall model** is a **linear, sequential** approach to the SDLC. Each phase must be completed before the next one begins, like water falling down steps. There is no easy way to go back to a previous phase.



The Waterfall model: each phase "falls" into the next, in one direction only.

When to use it

- Requirements are clear, fixed and well understood.
- The project is small to medium in size.
- The technology is stable (not new or experimental).

Advantages (Pros)	Disadvantages (Cons)
Simple to understand and manage.	Not flexible to changes.
Well documented at every step.	Testing happens very late in the process.

WORKED EXAMPLE

Case study: Al-Noor Library Management System. The textbook uses this system as a Waterfall case study because the requirements were **100% fixed before coding began**.

Step 1: The librarian listed exactly what was needed: add books, issue books, return books, calculate fines. Nothing would change.

Step 2: Because the list was fixed, the team could finish the full design, then all the coding, then all the testing, in order.

Step 3: Every phase produced a document, so the library has a complete record for future maintenance.

REMEMBER IT

Waterfall = water only flows downwards. Once water has fallen it cannot climb back up — once a phase is finished you do not go back.

COMMON MISTAKE

Do not choose Waterfall for a project where the customer is still unsure what they want. A late change means going back through design, coding and testing, which is slow and costly.

اردو میں سمجھیں

وائرفال (آبشار) ماڈل ایک سیدھا اور ترتیب وار طریقہ ہے؛ ہر مرحلہ مکمل ہونے کے بعد ہی اگلا شروع ہوتا ہے اور پیچھے جانا مشکل ہوتا ہے۔ یہ تب بہتر ہے جب ضروریات واضح اور طے شدہ ہوں، جیسے النور لائبریری مینجمنٹ سسٹم میں۔ یہ سمجھنے میں آسان ہے مگر تبدیلیوں کے لیے لچکدار نہیں اور ٹیسٹنگ دیر سے ہوتی ہے۔

EXAM TIP

Expect "Explain the Waterfall model with its advantages and disadvantages" or "When should the Waterfall model be used?" Draw the step diagram, list the three "when to use" conditions, and give at least two pros and two cons. Mention the Al-Noor Library case study as your example.

CHECK YOURSELF

1. Why is the Waterfall model called "sequential"?
2. Why did the Al-Noor Library system use Waterfall?

Answers: 1. Because each phase must be completed before the next begins, in a fixed order. 2. Because its requirements were 100% fixed before coding began.

1.2.4 The Agile Model ★ Golden

LEARN IT

The **Agile model** is an **iterative** approach: the software is built in small, repeated cycles called **sprints** (usually 2–4 weeks long). Each sprint plans, builds and tests a small part of the software, and then shows it to the customer. **Customer feedback is gathered continuously** and used in the next sprint.

Plan → Design → Build → Test → Review & feedback → Next sprint ↻

When to use it

- Requirements are unclear or may change.
- The customer wants frequent progress updates (early delivery).
- The project is complex or innovative.

Advantages (Pros)	Disadvantages (Cons)
Highly flexible to changes.	Less documentation.
Early delivery of working software.	Unpredictable end-time and cost.

WORKED EXAMPLE

Case study: Al-Noor Student Mobile Application. This app used Agile because teachers, parents and students **constantly requested new features** during development.

Sprint 1 (2 weeks): build login and the timetable screen. Show it to users.

Feedback: parents ask, "Can we see attendance too?"

Sprint 2: add attendance. Teachers now ask for homework notifications.

Sprint 3: add notifications. Each sprint delivers a working app, and new ideas are welcomed instead of causing a restart.

REMEMBER IT

Agile = agile athlete: quick and able to change direction. Think of making a pizza where the customer tastes each slice and you adjust the next one.

COMMON MISTAKE

"Less documentation" does not mean "no documentation" or "no planning". Agile teams still plan every sprint; they simply value working software over long documents.

اردو میں سمجھیں

ایجائل (Agile) ماڈل میں سافٹ ویئر چھوٹے چھوٹے ادوار میں بنتا ہے جنہیں اسپرنٹ (Sprint) کہتے ہیں، جو عموماً دو سے چار ہفتے کے ہوتے ہیں۔ ہر اسپرنٹ کے بعد گاہک کی رائے لی جاتی ہے۔ یہ تبدیلیوں کے لیے بہت لچکدار ہے اور جلد کام کرنے والا سافٹ ویئر دیتا ہے، مگر اس میں دستاویزات کم ہوتی ہیں اور وقت و لاگت کا اندازہ مشکل ہوتا ہے۔ النور اسٹوڈنٹ موبائل ایپ اس کی مثال ہے۔

EXAM TIP

Questions ask you to define Agile, explain sprints, list pros/cons, or say when to use it. Always mention the words "iterative", "sprints (2–4 weeks)" and "continuous customer feedback". Use the Al-Noor Student Mobile App as your case study.

CHECK YOURSELF

1. What is a sprint and how long does it usually last?
2. Give one disadvantage of Agile.

Answers: 1. A short development cycle that delivers a working part of the software; usually 2–4 weeks. 2. Less documentation, or unpredictable end-time and cost.

1.2.5 – 1.2.7 Waterfall vs Agile: Comparison and Case Studies

LEARN IT

Both models follow the same SDLC phases. The difference is **how** they move through them: Waterfall goes through each phase **once**, in a straight line; Agile goes through all phases **many times**, in small loops.

Dimension	Waterfall model	Agile model
Flexibility	Rigid. Hard to make changes later.	Very flexible. Adapts to changes easily in the next sprint.
Customer involvement	Only at the start (requirements) and the end (delivery).	Continuous throughout the project (after every sprint).
Delivery of software	Single delivery at the very end.	Frequent, partial deliveries after every sprint.
Testing	Done late, in a dedicated testing phase.	Continuous testing within every sprint.
Documentation	Heavy documentation required.	Light documentation. Focus is on working software.

WORKED EXAMPLE

Choosing a model — think like the textbook.

Case A: Al-Noor Library Management System. Requirements fixed 100% before coding → no changes expected → **Waterfall**.

Case B: Al-Noor Student Mobile Application. Teachers, parents and students keep asking for features → requirements change → **Agile**.

Practice case: A bank wants a salary-calculation program for a fixed government formula that will not change. Requirements are clear and fixed, the project is medium-sized, and the technology is stable → **Waterfall**.

Practice case: A start-up wants a new food-delivery app and is not sure which features customers will like. Requirements are unclear and the project is innovative → **Agile**.

REMEMBER IT

One question decides it: **"Will the requirements change?"** No → Waterfall. Yes / not sure → Agile.

COMMON MISTAKE

Do not say one model is always "better". Each suits a different situation. In exams, justify your choice using the project's features (fixed vs changing requirements, size, customer involvement).

اردو میں سمجھیں

وائرفال ماڈل سخت (Rigid) ہے، گاہک صرف شروع اور آخر میں شامل ہوتا ہے، سافٹ ویئر ایک ہی بار آخر میں ملتا ہے اور دستاویزات زیادہ ہوتی ہیں۔ ایجائل ماڈل لچکدار ہے، گاہک ہر اسپرنٹ کے بعد شامل ہوتا ہے، سافٹ ویئر بار بار حصوں میں ملتا ہے اور ٹیسٹنگ مسلسل ہوتی ہے۔ اگر ضروریات طے شدہ ہوں تو وائرفال اور اگر بدلتی رہیں تو ایجائل منتخب کریں۔

CHECK YOURSELF

1. In which model is the customer involved only at the start and the end?
2. Which model delivers software in frequent, partial deliveries?
3. A school wants an exam-result system with rules that never change. Which model?

Answers: 1. Waterfall. 2. Agile. 3. Waterfall (fixed, clear requirements).

1.3.1 – 1.3.4 Communication Models and the OSI 7-Layer Model

★ Golden

LEARN IT

A **communication model** explains how data travels from one device to another through a network. Because sending data is complicated, the job is split into **layers**; each layer does one part of the job and passes the data to the next layer.

The **OSI (Open Systems Interconnection) model** was developed by **ISO** (International Organization for Standardization) in **1984**. It divides communication into **7 layers**. When you send data, it goes **down** the layers on your device (7 → 1), travels as bits over the cable or air, and goes **up** the layers on the receiving device (1 → 7).



The 7 layers of the OSI model with the mnemonic "Please Do Not Throw Sausage Pizza Away".

REMEMBER IT

Mnemonic (bottom to top, 1 → 7): "Please Do Not Throw Sausage Pizza Away" = Physical, Data Link, Network, Transport, Session, Presentation, Application. (Top to bottom you can use "All People Seem To Need Data Processing".)

OSI layers — top layers (7 to 4)

Layer	Function	Protocols / examples
7. Application	Provides network services directly to user applications (browser, email program).	HTTP, HTTPS, FTP, SMTP, DNS
6. Presentation	Data translation, encryption (security) and compression.	SSL/TLS, JPEG, MP3, ASCII
5. Session	Opens, maintains and closes communication sessions.	NetBIOS, RPC
4. Transport	Reliable end-to-end delivery, error checking, port numbering.	TCP (reliable), UDP (fast)

OSI layers — bottom layers (3 to 1)

Layer	Function	Protocols / hardware
3. Network	Logical addressing (IP addresses) and routing packets across networks.	IP, ICMP (Ping), Routers
2. Data Link	Physical addressing (MAC addresses) and error detection in frames.	Ethernet, MAC, Switches
1. Physical	Transmits raw bits (0, 1) over cables or wireless media.	Cables, Hubs, Wi-Fi radio

TCP vs UDP at the Transport layer

	TCP	UDP
Main feature	Reliable — checks that all data arrives correctly and in order.	Fast — sends data without waiting for checks.
Use when	Accuracy matters.	Speed matters more than perfection.
Examples	Web browsing, emails	Video streaming, online gaming

WORKED EXAMPLE**Tracing an email through the OSI layers (sender side).**

Layer 7 (Application): you press Send; your email program uses **SMTP**.

Layer 6 (Presentation): the text is encoded (ASCII) and may be encrypted with TLS.

Layer 5 (Session): a session is opened with the mail server.

Layer 4 (Transport): **TCP** breaks the data into segments and numbers them, so nothing is lost (email needs accuracy).

Layer 3 (Network): each packet gets the **IP address** of the destination; **routers** choose the path.

Layer 2 (Data Link): packets are put in **frames** with **MAC addresses**; a **switch** forwards them on the local network.

Layer 1 (Physical): the frames are sent as raw **bits** over the cable or Wi-Fi radio.

At the receiver, the same steps happen in reverse order (1 → 7).

COMMON MISTAKE

Do not mix up the two addresses: **IP address** = logical address = **Network layer (3)**; **MAC address** = physical address = **Data Link layer (2)**. Also, routers work at layer 3, switches at layer 2 and hubs at layer 1.

اردو میں سمجھیں

کمپونیکیشن ماڈل بتاتا ہے کہ ڈیٹا نیٹ ورک کے ذریعے ایک ڈیوائس سے دوسری تک کیسے پہنچتا ہے۔ او ایس آئی (OSI) ماڈل 1984 میں آئی ایس او (ISO) نے بنایا اور اس کی سات تہیں (Layers) ہیں: فزیکل، ڈیٹا لنک، نیٹ ورک، ٹرانسپورٹ، سیشن، پریزنٹیشن اور ایپلی کیشن۔ ہر تہ کا الگ کام ہے، جیسے نیٹ ورک تہ IP ایڈریس اور روٹنگ سنبھالتی ہے۔ TCP درستگی کے لیے اور UDP رفتار کے لیے استعمال ہوتا ہے۔

EXAM TIP

A guaranteed long question: "Explain the seven layers of the OSI model." Draw the 7-layer stack (numbered, layer 7 on top), then write one function and one or two protocols/devices for each layer. MCQs often ask: "Which layer handles routing?" (Network), "Which layer does encryption?" (Presentation), "Who developed OSI and when?" (ISO, 1984), "Which protocol is used for video streaming?" (UDP).

CHECK YOURSELF

1. Which OSI layer is responsible for encryption and compression?
2. At which layer do switches and MAC addresses work?
3. Should online gaming use TCP or UDP? Why?

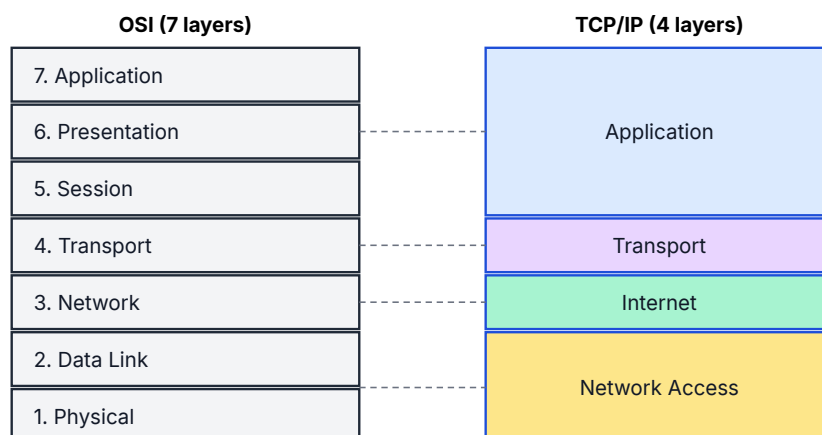
Answers: 1. Presentation layer (6). 2. Data Link layer (2). 3. UDP, because speed matters more than perfect accuracy.

1.3.5 – 1.3.6 TCP/IP Model vs OSI Model

LEARN IT

The OSI model is mainly a **reference** model used for teaching and design. The **TCP/IP model** is the **practical model used on the real Internet**. It condenses the 7 OSI layers into just **4 layers**.

TCP/IP model (4 layers)	Corresponds to OSI layers
1. Application	Application, Presentation, Session (7, 6, 5)
2. Transport	Transport (4)
3. Internet	Network (3)
4. Network Access	Data Link, Physical (2, 1)



How the 7 OSI layers map onto the 4 TCP/IP layers.

Point	OSI model	TCP/IP model
Number of layers	7	4
Use	Reference / teaching model	Practical model used on the real Internet
Top layers	Separate Application, Presentation and Session layers	All three combined into one Application layer
Bottom layers	Separate Data Link and Physical layers	Combined into one Network Access layer

WORKED EXAMPLE

Where does each protocol/device go in TCP/IP?

HTTP (OSI layer 7) → TCP/IP **Application**.

SSL/TLS (OSI layer 6) → TCP/IP **Application** (layers 7, 6, 5 are merged).

UDP (OSI layer 4) → TCP/IP **Transport**.

IP and routers (OSI layer 3) → TCP/IP **Internet**.

Ethernet, switches, cables (OSI layers 2 and 1) → TCP/IP **Network Access**.

REMEMBER IT

Count it as **3-1-1-2**: the TCP/IP Application layer takes 3 OSI layers, Transport takes 1, Internet takes 1, Network Access takes 2. Total $3+1+1+2 = 7$.

COMMON MISTAKE

The OSI "Network" layer (3) matches the TCP/IP "**Internet**" layer, not the TCP/IP "Network Access" layer. The similar names confuse many students.

اردو میں سمجھیں

ٹی سی پی/آئی پی (TCP/IP) ماڈل حقیقی انٹرنیٹ پر استعمال ہونے والا عملی ماڈل ہے اور اس کی صرف چار تہیں ہیں۔ اس کی ایپلی کیشن تہ OSI کی 7، 6 اور 5 تہوں کے برابر ہے، ٹرانسپورٹ تہ 4 کے، انٹرنیٹ تہ 3 کے، اور نیٹ ورک ایکسیس تہ 2 اور 1 کے برابر ہے۔

CHECK YOURSELF

1. How many layers does the TCP/IP model have?
2. Which TCP/IP layer corresponds to OSI layers 2 and 1?
3. Which TCP/IP layer contains the OSI Session layer?

Answers: 1. Four. 2. Network Access. 3. Application.

Chapter 1 Review

Chapter summary

- Discrete quantities are counted (students, cars); continuous quantities are measured and can take any value (temperature, speed). Stairs vs ramp.
- A digital system uses only two values, 0 (OFF/LOW/FALSE) and 1 (ON/HIGH/TRUE), called bits. It is reliable, accurate, easy to design and programmable.
- Analog signals change smoothly (sine wave) and are affected by noise; digital signals jump between HIGH and LOW (square wave) and resist noise.
- Boolean algebra, by George Boole, works with only TRUE (1) and FALSE (0) using variables, operations and expressions.
- AND (·) gives 1 only when all inputs are 1; OR (+) gives 1 when any input is 1; NOT (') inverts its single input (unary).
- A truth table lists all 2^n input combinations in binary order with their outputs.
- Logic gates are the building blocks of digital systems: AND, OR, NOT (basic); NAND, NOR (universal); XOR, XNOR (advanced).
- NAND and NOR are universal: any circuit can be built using only one of them.
- XOR gives 1 when inputs are different; XNOR gives 1 when inputs are the same.
- Order of precedence: Parentheses → NOT → AND → OR. A logic diagram is drawn in the same order.
- Minterms (SOP) come from rows with output 1; maxterms (POS) come from rows with output 0.
- K-maps use Gray code (00, 01, 11, 10); group 1s in sizes 1, 2, 4, 8, as large as possible; overlap and wrap-around are allowed; changing variables are eliminated.
- LogiSim Evolution simulates circuits: input pins are squares, output pins circles; the Poke Tool toggles inputs; bright green = 1, dark green = 0, blue = unconnected, red = error.
- SDLC has 6 phases: Requirement Analysis (SRS), Design, Implementation, Testing (black box/white box), Deployment (direct/phased/pilot), Maintenance.
- Waterfall is linear and suits fixed requirements (AI-Noor Library); Agile is iterative with 2–4 week sprints and suits changing requirements (AI-Noor Student App).
- The OSI model (ISO, 1984) has 7 layers — "Please Do Not Throw Sausage Pizza Away"; the practical TCP/IP model has 4 layers: Application, Transport, Internet, Network Access.

Key terms

Term	Meaning	اردو
Discrete quantity	A countable quantity with separate, fixed values.	منفصل مقدار
Continuous quantity	A quantity that can take any value within a range.	مسلسل مقدار
Bit	A binary digit, 0 or 1.	بائری ہندسہ
Digital system	An electronic system that uses only the values 0 and 1.	ڈیجیٹل نظام
Analog signal	A continuous signal that changes smoothly, like a sine wave.	اینالوگ سگنل (مسلسل اشارہ)
Digital signal	A signal with only two levels, HIGH and LOW (square wave).	ڈیجیٹل سگنل
Noise	Unwanted interference that disturbs a signal.	شور (مداخلت)
Boolean algebra	Algebra of two values, TRUE (1) and FALSE (0), developed by George Boole.	بولین الجبرا
Truth table	A table of all input combinations and their outputs.	ٹروٹھ ٹیبل (صداقت جدول)
Logic gate	An electronic circuit that performs a Boolean operation.	منطقی گیٹ
Universal gate	A gate (NAND or NOR) from which any circuit can be built.	عالمگیر گیٹ
Unary operation	An operation with only one operand, e.g. NOT.	یک عددی عمل
Minterm	A product (AND) of all variables for a row with output 1.	منٹرم
Maxterm	A sum (OR) of all variables for a row with output 0.	میکس ٹرم
SOP	Sum of Products: minterms joined with OR.	حاصل ضرب کا مجموعہ
POS	Product of Sums: maxterms joined with AND.	مجموعوں کا حاصل ضرب
Karnaugh map (K-map)	A grid used to simplify Boolean expressions by grouping 1s.	کارنو نقشہ
Gray code	An ordering where neighbouring values differ by only one bit.	گرے کوڈ
Simulation	Running a computer model of a circuit to test it.	نقلی عمل (سمیولیشن)

Term	Meaning	اردو
SDLC	Software Development Life Cycle: the 6-phase process for building software.	سافٹ ویئر کی تیاری کا دورِ حیات
SRS	Software Requirements Specification: document listing what the software must do.	سافٹ ویئر ضروریات کی دستاویز
Black box testing	Testing inputs and outputs without seeing the code.	بلیک باکس جانچ
White box testing	Testing the internal logic while seeing the code.	وائٹ باکس جانچ
Waterfall model	A linear SDLC model; each phase is finished before the next.	آبشار ماڈل
Agile model	An iterative SDLC model using short sprints and constant feedback.	ایجائل (لچکدار) ماڈل
Sprint	A short development cycle in Agile, usually 2–4 weeks.	اسپرینٹ (مختصر ترقیاتی دور)
Protocol	A set of rules for communication, e.g. HTTP, TCP.	پروٹوکول (ضوابطِ ابلاغ)
OSI model	ISO's 7-layer reference model of network communication (1984).	او ایس آئی ماڈل
TCP/IP model	The practical 4-layer model used on the Internet.	ٹی سی پی/آئی پی ماڈل

Practice: multiple choice

1. Which of the following is a continuous quantity?
 - a. Number of students in a class
 - b. Number of cars in a parking lot
 - c. Temperature of water
 - d. Number of books on a shelf
2. In a digital system, the value 1 represents:
 - a. OFF
 - b. LOW
 - c. FALSE
 - d. ON / HIGH / TRUE
3. A square wave represents a:
 - a. Digital signal
 - b. Analog signal
 - c. Sine wave
 - d. Noise signal
4. Boolean algebra was developed by:
 - a. Charles Babbage
 - b. George Boole
 - c. Alan Turing
 - d. Maurice Karnaugh
5. How many rows does a truth table with 4 inputs have?
 - a. 4
 - b. 8
 - c. 16
 - d. 32
6. Which gate gives output 1 only when all inputs are 1?
 - a. OR
 - b. AND
 - c. NOR
 - d. XOR
7. Which pair of gates are called universal gates?
 - a. AND and OR
 - b. XOR and XNOR
 - c. NOT and AND
 - d. NAND and NOR
8. What is the output of a NOR gate when $A = 0$ and $B = 0$?
 - a. 1
 - b. 0
 - c. Unknown
 - d. Depends on the gate size
9. An XOR gate gives output 1 when:
 - a. Both inputs are 1
 - b. Both inputs are 0
 - c. The inputs are different
 - d. The inputs are the same
10. After parentheses, which operation has the highest precedence?
 - a. OR
 - b. NOT
 - c. AND
 - d. XOR
11. The minterm for the row $A = 1, B = 0$ is:
 - a. AB'
 - b. $A'B$
 - c. $A + B'$
 - d. $A' + B$
12. The columns of a 3-variable K-map (BC) are written in the order:
 - a. 00, 01, 10, 11
 - b. 00, 10, 01, 11
 - c. 11, 10, 01, 00
 - d. 00, 01, 11, 10
13. In black box testing, the tester:
 - a. Reads the source code line by line
 - b. Does not see the code and tests inputs/outputs
 - c. Is always the developer
 - d. Tests only the internal structure
14. Which OSI layer handles logical (IP) addressing and routing?
 - a. Data Link
 - b. Transport
 - c. Network
 - d. Session
15. The TCP/IP Network Access layer corresponds to which OSI layers?
 - a. 7, 6 and 5
 - b. 4
 - c. 3
 - d. 2 and 1

Practice: short questions

1. Define discrete and continuous quantities. Give two examples of each.
2. State four advantages of digital systems.
3. Differentiate between analog and digital signals (any three points).
4. Why is NOT called a unary operation? Write its truth table.
5. How many rows are needed in a truth table for 3 inputs? List all the input combinations.
6. Why are NAND and NOR called universal gates? Show how a NAND gate can be used as a NOT gate.
7. A 2-input truth table has $Y = 1$ only for rows $A=0, B=1$ and $A=1, B=0$. Write the SOP expression and name the gate.
8. Name the deliverable of each SDLC phase.
9. Differentiate between black box testing and white box testing.
10. When should TCP be used and when should UDP be used? Give one example of each.

Practice: long / problem questions

1. Draw the symbol, write the Boolean expression and give the truth table for each of these gates: AND, OR, NOT, NAND, NOR, XOR and XNOR.
2. A function is given as $F(A, B, C) = \sum m(1, 3, 5, 6, 7)$. (a) Make its truth table. (b) Simplify it using a 3-variable K-map, explaining each group. (c) Draw the logic diagram of the simplified expression.
3. For $Y = A \cdot B' + C$: (a) state the order in which the operations are done; (b) make the complete truth table; (c) draw the logic diagram.
4. Explain the six phases of the Software Development Life Cycle with the activities and deliverable of each phase. Use the house-building analogy.
5. Compare the Waterfall and Agile models under five headings. Explain why the Al-Noor Library Management System used Waterfall and the Al-Noor Student Mobile Application used Agile.
6. Explain the seven layers of the OSI model with the function and one protocol/device for each layer. Then show how they map onto the four layers of the TCP/IP model.

Answer key

Multiple choice

1. c
2. d
3. a
4. b
5. c
6. b
7. d
8. a
9. c
10. b
11. a
12. d
13. b
14. c
15. d

Short questions — model answers

1. Discrete quantities are separate, countable items with specific values (number of students in a classroom, number of cars in a parking lot). Continuous quantities can take any value within a range and flow without breaks (temperature of water, speed of a moving car).
2. Reliable (less affected by noise and interference); accurate (no loss of quality when copying); easy to design (only two states, 0 and 1); programmable (easily controlled by software).
3. Nature: analog is a continuous flow; digital has discrete values (bits, HIGH/LOW). Shape: analog is a sine wave; digital is a square wave. Reliability: analog is low (affected by noise); digital is high (resistant to noise). Examples: human voice vs computer data.
4. NOT works on only one operand (one input), so it is unary. Truth table: $A = 0 \rightarrow Y = 1$; $A = 1 \rightarrow Y = 0$.
5. $2^3 = 8$ rows: 000, 001, 010, 011, 100, 101, 110, 111.
6. Any logic circuit can be built using only NAND gates, or only NOR gates. NAND as NOT: join both inputs and feed A: $Y = (A \cdot A)' = A'$. Check: $A = 0 \rightarrow 1$, $A = 1 \rightarrow 0$.
7. Minterms: $A'B$ (row 01) and AB' (row 10). $Y = A'B + AB'$. This is the XOR gate, $Y = A \oplus B$.
8. Requirement Analysis $\rightarrow$ SRS document; System Design $\rightarrow$ Design Document; Implementation $\rightarrow$ Working Source Code; Testing $\rightarrow$ Test Reports / bug-free software; Deployment $\rightarrow$ Live Software; Maintenance $\rightarrow$ Patches / Updated Versions.
9. Black box: tester does not see the code, tests functionality (inputs/outputs), done by dedicated testers. White box: tester can see the code, tests internal logic/structure, done by developers.
10. Use TCP when accuracy matters, e.g. web browsing or email. Use UDP when speed matters more than perfection, e.g. video streaming or online gaming.

Long questions — model answers

- Symbols as drawn in 1.1.7 (AND = D shape; OR = curved back, pointed front; NOT = triangle + bubble; NAND = AND + bubble; NOR = OR + bubble; XOR = OR with extra back curve; XNOR = XOR + bubble). Expressions: AND $Y = A \cdot B$; OR $Y = A + B$; NOT $Y = A'$; NAND $Y = (A \cdot B)'$; NOR $Y = (A + B)'$; XOR $Y = A \oplus B$; XNOR $Y = (A \oplus B)'$. Outputs for inputs 00, 01, 10, 11: AND 0,0,0,1; OR 0,1,1,1; NAND 1,1,1,0; NOR 1,0,0,0; XOR 0,1,1,0; XNOR 1,0,0,1. NOT: 0 $\rightarrow$ 1, 1 $\rightarrow$ 0.
- (a) Truth table ($ABC \rightarrow F$): 000 $\rightarrow$ 0, 001 $\rightarrow$ 1, 010 $\rightarrow$ 0, 011 $\rightarrow$ 1, 100 $\rightarrow$ 0, 101 $\rightarrow$ 1, 110 $\rightarrow$ 1, 111 $\rightarrow$ 1.

(b) K-map:

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	0	1	1	0
A = 1	0	1	1	1

Group 1 (size 4): the two middle columns, m_1, m_3, m_5, m_7 . A changes and B changes; C = 1 stays $\rightarrow$ **C**. Group 2 (size 2): m_7 and m_6 in row A = 1. C changes; A = 1, B = 1 stay $\rightarrow$ **AB** (m_7 is shared — overlapping is allowed). **F = C + AB**. Check: C + AB is 1 for rows 1, 3, 5, 7 (C = 1) and 6, 7 (AB = 1) = {1, 3, 5, 6, 7}. ✓

(c) Logic diagram: an AND gate with inputs A and B gives AB; an OR gate takes AB and C and gives F.

- (a) First NOT (B'), then AND ($A \cdot B'$), then OR ($+ C$).

(b) Truth table:

A	B	C	B'	A·B'	Y
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1

(c) Logic diagram: input B goes through a NOT gate to give B' . A and B' go into an AND gate to give $A \cdot B'$. $A \cdot B'$ and C go into an OR gate to give Y. (3 gates.)

- Requirement Analysis: meet clients, understand needs, list features; most important phase; deliverable SRS (house: asking the family what rooms they need). System Design: high-level design (architecture), low-level design (database, UI screens), flowcharts; deliverable Design Document (architect draws blueprints). Implementation: write code in Python, Java etc. from the design; deliverable Working Source Code (builders lay bricks). Testing: find and fix bugs using black box and white box testing; deliverable Test Reports / bug-free software (inspector checks the house). Deployment: release to end-users by direct, phased or pilot release; deliverable Live Software (family moves in). Maintenance: support, updates, fixing post-release issues; deliverable Patches / Updated Versions (repairs and extensions).
- Comparison: Flexibility — Waterfall rigid, Agile very flexible. Customer involvement — Waterfall only at start and end, Agile continuous after every sprint. Delivery — Waterfall single delivery at the end, Agile frequent partial deliveries. Testing — Waterfall late in a dedicated phase, Agile continuous in every sprint. Documentation — Waterfall heavy, Agile light. The Al-Noor Library system used Waterfall because its requirements were 100% fixed before coding began, so a simple, well-documented linear process fitted. The Al-Noor Student App used Agile because teachers, parents and students kept requesting new features, so short sprints with continuous feedback were needed.
- OSI (ISO, 1984), top to bottom: 7 Application — network services to user applications (HTTP, HTTPS, FTP, SMTP, DNS); 6 Presentation — translation, encryption, compression (SSL/TLS, JPEG, MP3, ASCII); 5 Session — opens, maintains, closes sessions (NetBIOS, RPC); 4 Transport — reliable end-to-end delivery, error checking, port numbers (TCP, UDP); 3 Network — logical IP addressing and routing (IP, ICMP, routers); 2 Data Link — physical MAC addressing and error detection in frames (Ethernet, switches); 1 Physical — raw bits over cables/wireless (cables, hubs, Wi-Fi radio). Mnemonic bottom to top: "Please Do Not Throw Sausage Pizza Away". TCP/IP mapping: Application = OSI 7, 6, 5; Transport = 4; Internet = 3; Network Access = 2, 1.

Self-assessment: I can...

- ☐ ...tell discrete from continuous quantities and analog from digital signals, and draw both waves.
- ☐ ...evaluate AND, OR and NOT expressions and follow the order of precedence.
- ☐ ...draw all seven gate symbols and write their truth tables from memory.
- ☐ ...write the SOP (minterms) and POS (maxterms) from a truth table.
- ☐ ...simplify expressions with 2- and 3-variable K-maps, including overlap and wrap-around groups.
- ☐ ...explain the six SDLC phases and choose between Waterfall and Agile for a project.
- ☐ ...list four advantages of digital systems.
- ☐ ...build a complete truth table for 2 or 3 inputs without missing a row.
- ☐ ...explain why NAND and NOR are universal gates.
- ☐ ...draw a logic diagram from an expression.
- ☐ ...build and test a simple circuit in LogiSim and read the wire colours.
- ☐ ...name the 7 OSI layers in order, state their functions, and map them to the 4 TCP/IP layers.